School of Computing and Mathematics sciences

CO7201 Individual Project

Interim Report

Placement management web application

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DECLARATION

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Date: 27/03/2026

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I Introduction

The **Placement Management System** is being developed to simplify and digitalize the student placements in universities. This project is building on a full-stack application that manages placement activities and improves structured data and analytics. This report tells the progress made, including completed work, current development, and the next steps ahead

Status of Requirements

Each paragraph in this section states the previously mentioned essential requirements and then details their current status. The requirements previously listed as recommended and optional have not been started yet, and thus have not been discussed in this report

II Project Overview

Since the submission of the preliminary report, progress has been made in both frontend and backend development. The project has moved from **planning** and **design phase** to **implementation phase.**

The development has been divided into two major parts:

- Frontend Development
- Backend Development

At this stage the frontend for all major modules has been completed, while the backend is developed, partially implemented and actively ongoing.

II.1 Completed Tasks

Frontend Development

A major portion of the design and implementation of the user interface is completed. The aim is to ensure a clean, responsive, and user-friendly experience across all modules.

- The Homepage has been fully developed, providing a clear entry point to the application with navigation. It gives an overview of the system and lets the user jump to a relevant area depending on the role.
- The admin dashboard has been implemented. The module contains the students, tutors, departments, and companies. This interface is designed to show data in an organized manner.

- Interfaces for the students, tutors, and companies were developed. and registrations for students and companies were designed and developed.
- Along with the frontend, the backend is also developed. A login authentication system has been developed to secure access to the platform.
- Implementation of tutor management is completed; similarly, department management functionality has been developed. These modules are the main part of the system's core functionality.
- Essential things like Enums have been defined to support backend logic. Which include representations for roles, statues, and other system-specific classifications, developing these structures early will help to maintain consistency.

II.2 Ongoing Tasks

Currently, we are working towards developing integration of the backend and frontend. The user interface is mostly completed.

- Work is in progress with the remaining back-end modules, which include student management, company management, and application processing. These modules need to be designed carefully to make sure that all system entities are connected and that data flows between them.
- At the same time, we also actively worked on API development, i.e., provided endpoints to let the frontend communicate with the backend.
- Another important ongoing process is frontend-backend integration. The static user interfaces are slowly being connected to real backend data so that users can interact with the system in a dynamic way.
- Also working on the system architecture, which makes the code easier to read, makes sure it can be reused, and keeps the layers of the application separate.

II.3. Pending Task

There are still some important things missing from the project that are needed for the system to work properly and for the final application to be delivered.

- The immediate priority task is to complete the remaining backend functionalities. Modules like student management, company interaction, and the application process are still pending.
- Another issue is the full implementation of API integration and data mapping. This includes setting up relations between entities. It includes smooth communication between frontend and backend components. Handling data efficiently and maintaining consistency across the system.
- A feature that is intended to be added to the project in large stages is a prediction system.
- At the end, system testing and validation remain pending. This helps to fix the bugs and ensure system stability and performance.

III Changes Implemented

After getting feedback on the first report, changes have been made to make the project clearer and better

First, the link between how the project is technically carried out and its goals has gotten better. At this point in the project, the development of the frontend and backend components has been better aligned with the system requirements to make sure that each feature helps the Placement Management System reach its main goal.

The time plan has been improved to make it more realistic and organized. A detailed Gantt chart has been made to better show the timeline and keep track of progress.

Another important change is that there is now a clear way. Now it includes testing strategies like integration testing and performance validations to make sure the system is reliable. The background research has been more closely linked to the project implementation. The research is now shown in design choices and system architecture instead of listing ideas separately.

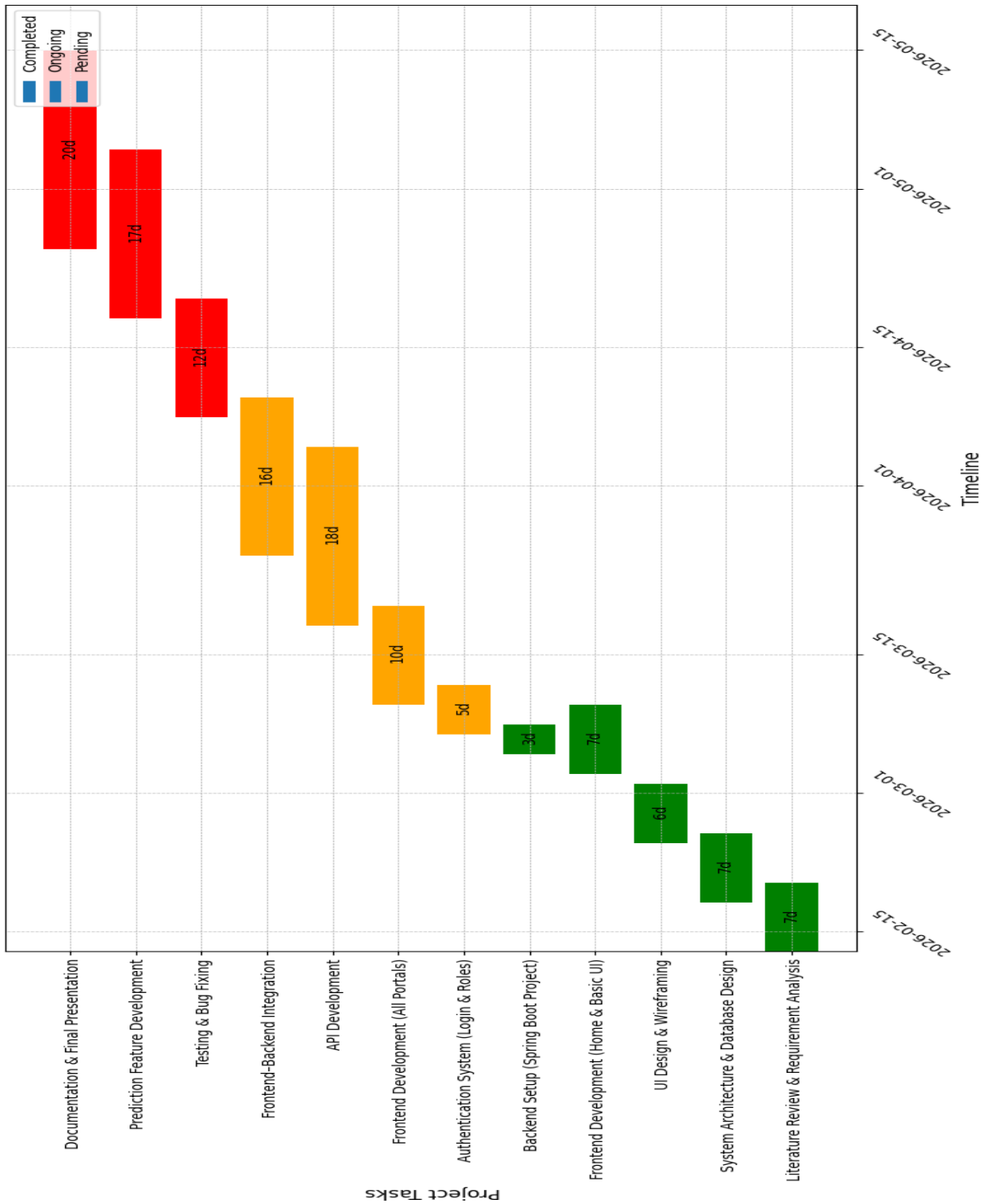
IV Challenges

Several challenges were encountered during the development of the project, particularly during the **implementation** and **integration** phases.

- One of the biggest problems was making sure that the frontend and backend parts worked together. Even though the user interface was finished first, connecting it to the backend service required careful management of the API's data flow and error handling.
- Another issue was how to make the backend work for different types of users. The system design was complex because each role needed different functions, data models, and relationships.
- Managing time was also hard, especially when I had to work on more than one module at the same time. It takes longer than planned because of the debugging and integration problems.
- Adding features like the prediction system requires more research and knowledge to implement it. This feature needs data that is set up correctly, so it will be added later in the project.

V. Time Plan

The project has been planned in a structured manner so that the important stages are developed. Tasks have been assigned to specific weeks to ensure a balanced workload. Progress has been scheduled. All the remaining work has been planned to be completed before the final submission.


Completed
Ongoing
Pending
20d
17d
12d
16d
18d
10d
5d
3d
7d
6d
7d
7d
Documentation & Final Presentation
Prediction Feature Development
Testing & Bug Fixing
Frontend-Backend Integration
API Development
Frontend Development (All Portals)
Authentication System (Login & Roles)
Backend Setup (Spring Boot Project)
Frontend Development (Home & Basic UI)
UI Design & Wireframing
System Architecture & Database Design
Literature Review & Requirement Analysis
Project Tasks
2026-02-15
2026-03-01
2026-03-15
2026-04-01
2026-04-15
2026-05-01
2026-05-15
Timeline